SIH/SUS 2024 — Final Analytical Plan & Scope

Final repository version — records executed analyses, retained supporting analyses, and explicit exclusions

This document supersedes the initial analytical planning document. It records the final analytical scope and prevents later project work from silently expanding the analysis.

1. Project Objective

To analyze patterns and variation in the hospital admissions in **SIH-SUS (Hospital Information System of Brazil's Unified Health System)** during 2024, examining patient characteristics, diagnostic case mix, healthcare resource utilization, length of stay, and hospitalization costs across demographic, facility, and geographic dimensions.

This framing is deliberately descriptive and variation-focused rather than performance/efficiency-focused. The dataset does not include clinical severity adjustment, staffing, or bed-capacity data needed to make defensible efficiency or quality-of-care claims — it supports patterns and variation, not causal or performance conclusions.

2. Dataset and unit of analysis

Dataset: SIH/SUS 2024 hospital admissions. Unit of analysis: N_AIH / admission-level record. The project describes the composition and characteristics of recorded hospital admissions; it does not estimate population-level admission rates because no population denominator is present.

3. Stakeholders

Primary: Health system / public health decision-makers

Ministry of Health, DATASUS/SUS planning functions, state and municipal health authorities, regional healthcare planners. The natural audience for system-level questions about hospitalization burden, resource use, and geographic variation — not single-facility operational decisions.

Secondary: Hospital network and facility planners

Relevant for facility- and specialty-level variables (CNES, ESPEC, DIAS_PERM, ICU/UTI utilization, VAL_TOT) — e.g., which facilities or specialties show consistently higher resource intensity.

Tertiary: Researchers and healthcare analysts

Relevant to this project's own portfolio positioning toward RWE/HEOR-style analysis — utilization, diagnosis, geography, and cost patterns are standard real-world-data analytical territory.

4. Final analytical framework

Question	Final focus	Status
Q1 — Who is using inpatient hospital services?	Age, sex, race/color and geography; selected data-quality/context checks.	Closed
Q2 — What clinical conditions account for hospital utilization?	Principal diagnosis/case mix, ICD-10 chapters, secondary-diagnosis burden, mortality and continuation/readmission context; clinical-resource relationships.	Closed
Q3 — How does resource utilization vary?	LOS, ICU/UCI utilization, mortality × utilization, continuation, specialty and selected demographic/clinical relationships.	Closed

Q4 — How do hospitalization costs vary with case mix and resource use?	VAL_TOT, cost components, LOS/cost, ICU/UCI cost components, specialty, diagnosis, secondary-diagnosis burden, continuation and selected financing context.	Closed

5. Q1 — Demographic composition

Executed: age, sex, race/color and geography, including age × sex and age × race/color summaries. Geography was interpreted cautiously using the available mappings.

Item	Final decision
IDADE + COD_IDADE	Retained; analytical age groups created.
SEXO	Retained and used.
RACA_COR	Retained and used with SIH/SUS documentation-based interpretation.
MUNIC_RES / state geography	Retained for descriptive geography; no population-rate inference.
UF_ZI / MUNIC_MOV	Not used as primary analytical dimensions.
NACIONAL / ETNIA	Not developed as major portfolio findings.
HOMONIMO	Not used as an analytical variable.
INSTRU / CBOR	Dropped; low incremental value relative to project objective.

6. Q2 — Clinical conditions and case mix

The clinical layer uses interpretable ICD-10 groupings rather than attempting to present every individual diagnosis code. Principal diagnosis defines case mix; secondary-diagnosis count provides a compact measure of documented diagnostic burden.

Item	Final decision
DIAG_PRINC	Core; ICD-10 chapter/case-mix analysis and cost relationships.
DIAG_SECUN + secondary fields	Core; secondary-diagnosis count and cost/resource relationships.
ICD-10 chapter grouping	Executed and retained.
Secondary diagnosis count × cost	Executed and retained; clear cost gradient.
MORTE	Executed and retained for mortality/utilization and mortality/cost context.
ICU utilization × diagnosis	Executed/retained; important clinical-resource relationship.
Continuation/readmission context	Executed using validated continuation logic; retained as supporting analysis.
TPDISEC	Dropped; not needed after diagnosis fields/count were established.
PROC_SOLIC / PROC_REA	Dropped; procedure-level analysis would expand scope without strengthening the main case-mix story.
INFEHOSP / CID_ASSO / CID_MORTE	Supporting candidates only; not developed into major portfolio analyses.

7. Q3 — Resource utilization

DIAS_PERM is the core LOS measure. ICU/UCI utilization and mortality were explicitly connected to resource use rather than treated only as standalone descriptive variables.

Item	Final decision
DIAS_PERM	Core LOS measure.

QT_DIARIAS	Exploratory only; strong overlap with DIAS_PERM, with most differences reflecting minor billing-day conventions. Excluded from core models.
MARCA_UTI	Retained for ICU utilization analysis.
MARCA_UCI	Retained as supporting UCI utilization information.
MORTE × ICU	Executed and retained.
ICU × ICD-10 chapter	Executed/retained.
Continuation × utilization	Executed and retained.
ESPEC	Retained; specialty-level utilization/cost intensity.
CNES	Dropped from facility-level comparison; facility conclusions would require stronger validation and add substantial complexity.
NATUREZA / NAT_JUR / GESTAO	Dropped from final analytical scope.
DT_INTER / DT_SAIDA	Used where required for LOS/temporal logic, not as standalone analytical themes.

8. Q4 — Cost, case mix and resource use

VAL_TOT is the primary expenditure outcome. Major components and selected resource/case-mix relationships are analyzed without expanding into an exhaustive financial-variable exercise.

Item	Final decision
VAL_TOT	Core cost outcome.
VAL_SH	Core supporting component; cost decomposition.
VAL_SP	Core supporting component; cost decomposition.
VAL_UTI	Supporting ICU-cost component; validated as genuinely variable. Add one focused cost-composition analysis.
VAL_UCI	Supporting UCI-cost component; validated as genuinely variable. Add one focused cost-composition analysis.
DIAS_PERM × cost	Core relationship.
ICU × cost	Core relationship.
ESPEC × cost	Executed and retained for meaningful/high-volume specialties.
Principal ICD-10 chapter × cost	Executed and retained.
Secondary diagnosis count × cost	Executed and retained.
Continuation × cost	Executed and retained as supporting analysis.
Cost per day	Retained only where DIAS_PERM > 0.
FINANC	Contextual only; not a major explanatory analysis.
FAEC_TP	Contextual only; incomplete lookup prevents strong substantive interpretation.
VAL_SADT	Dropped; not required for the core cost story.
US_TOT	Dropped; VAL_TOT is the established primary expenditure outcome.

9. Derived variables and rules

Derived item	Final status
age_group_pub	Executed and used.
ICD-10 chapter label	Executed and used.
secondary_diag_count	Executed and used.
LOS categories	Executed and used for cost gradient.
ICU flag	Executed and used.
Continuation flag	Executed using validated logic.
cost_per_day	Executed with DIAS_PERM > 0 denominator rule.
High-cost/percentile flag	Supporting only; no separate modelling exercise.

10. Explicit exclusions and rationale

These exclusions are deliberate scope decisions, not claims that the variables are intrinsically unimportant.

Excluded area	Reason
CNES facility-level analysis	Too granular for defensible facility-level interpretation within current scope.
QT_DIARIAS as core LOS	Largely redundant with DIAS_PERM; retained only as exploratory/appendix evidence.
FAEC_TP substantive analysis	Available lookup does not provide sufficient coverage for strong interpretation.
FINANC as major driver	Useful context, but not sufficiently rich for a central explanatory analysis.
VAL_SADT	Not needed for the main cost/case-mix/resource questions.
US_TOT	Not needed when VAL_TOT is the primary cost outcome.
Procedure-level analysis	Would expand scope without materially improving the core portfolio narrative.
Education/occupation/nationality as major themes	Low incremental value relative to the main analytical questions.

11. Interpretation and limitations

The analysis is descriptive and observational. Associations should not be presented as causal effects. In particular, higher expenditure among ICU admissions should be described as an observed association consistent with greater resource intensity, not as proof that ICU involvement caused the additional expenditure.

- Administrative/billing data are subject to coding conventions.
- DIAS_PERM and QT_DIARIAS are related but not identical measures.
- Extreme cost outliers are present and can affect means.
- Some external lookup/enrichment fields have incomplete documentation.
- Admission-level analysis does not by itself establish patient-level longitudinal outcomes.
- Facility-level conclusions were intentionally excluded from the current scope.

12. Final portfolio story

Layer	Representative focus
Q1 — Composition	Who is represented in the recorded admissions?
Q2 — Clinical case mix	Which diagnostic groups and diagnostic burden patterns are represented, and how do they relate to outcomes/resource use?
Q3 — Utilization	How do LOS, ICU/UCI use, specialty and continuation vary?
Q4 — Expenditure	How does expenditure vary with LOS, ICU/UCI involvement, specialty, diagnosis and diagnostic burden?
Analytical judgment	Which variables were deliberately retained, contextualized or excluded—and why?

13. Source/documentation note

Variable meanings and coding interpretations were based on SIH/SUS documentation and project-specific validation. Where documentation or lookup coverage was incomplete, the variable was restricted to contextual or exploratory use rather than being over-interpreted.